IBM Quantum Hub at
National Taiwan University
Quantum Practitioners webinar series

Noise characterization and error mitigation

April 1, 2026

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What you will learn today

Part 1

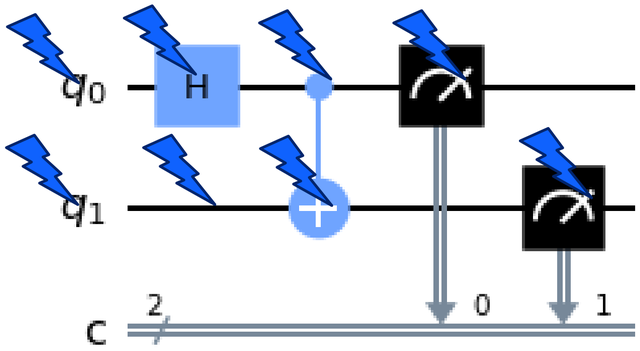
- Noise characterization, error suppression and mitigation
  - What is quantum noise/error
  - Error suppression and mitigation techniques
    - TREX (Twirled Readout Error eXtinction)
    - ZNE (Zero Noise Extrapolation)
    - PEA (Probabilistic Error Amplification)

Part 2

- Hands-on coding session
  - Noisy simulation without error mitigation
  - More realistic noisy simulation
  - Real quantum computation with error mitigation

Fighting noise

- Noise is everywhere:
    - Initialization
    - Gates (even in idling time)
    - Measurements



- **Noises** cause (computational) **errors**
- Errors prevent the realization of useful quantum computers

Quantum circuit optimization (last week) ➔ Reduce introduction of noise

Error mitigation (today) ➔ Fight remaining noise

Different sources of noise

We focus on **how noise affects computation (errors)**

Noise sources:

- Pulse distortion
- Material defects
- Heating
- Cosmic ray
- etc. (innumerable)

⋮

Noise in quantum computation

Classification of errors
(CS view)

Affected operations:

- **Coherent error**
- **Incoherent error**

} Gates

- **SPAM error** — Resets & measurements
  (SPAM: State Preparation And Measurement)
- **Sampling error** — Measurements

Engineering approach:

- Study the mechanism of noise
- Consider how to protect the device from the noise

Computer science (CS) approach: **Today**

- Study how noise affects computation
- Consider how to minimize the effect of noises (errors)

Coherent error (Unitary error)

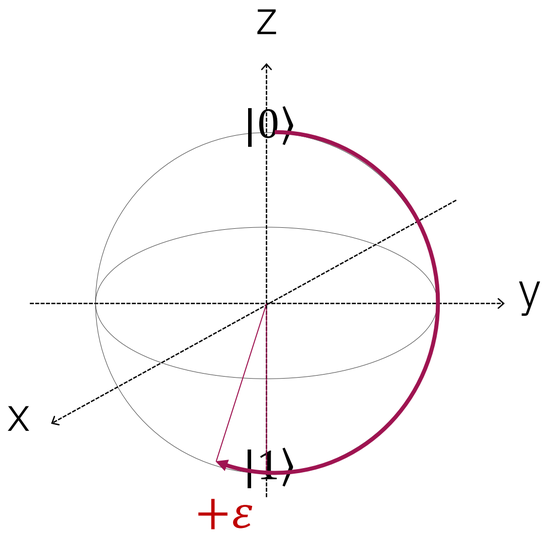
[Sources]
- Miscalibration (e.g. pulse amplitudes, qubit frequency)
- Unwanted interaction between qubits

[Characters]
- Unitary evolution, No change in purity (pure state → pure state)

[Measure]
- (Better calibration), Error mitigation/suppression

Ex) Miscalibration of an X gate

$$R_x(\pi) \xrightarrow{\text{Miscalibration}} R_x(\pi + \varepsilon)$$



Purity: $\mathrm{tr}(\rho^2)$

$$\underset{\text{mixed}}{\frac{1}{d}} \le \mathrm{tr}(\rho^2) \le \underset{\text{pure}}{1}$$

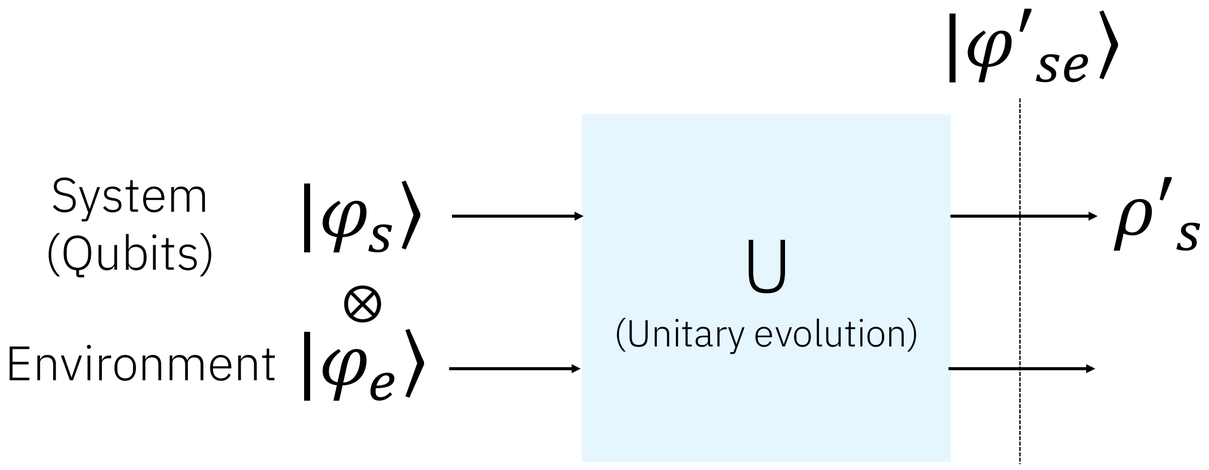
Miscalibration may cause over/under rotation errors

Incoherent error

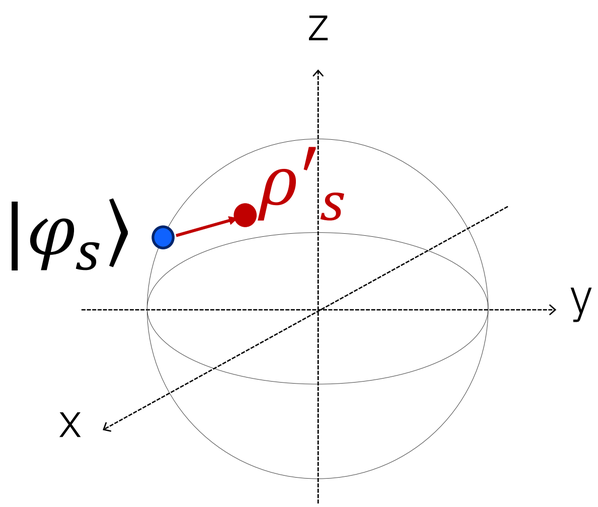
[Sources] • Entanglement (coupling) with environment (system is open)

[Characters] • Non-unitary, Loss of purity (pure state → mixed state)

[Measure] • Error mitigation



Quantum interaction as a whole



$$\rho'_s = \mathrm{tr}_e(|\varphi'_{se}\rangle\langle\varphi'_{se}|)$$

Partial trace over e (discard environment e, leave system s)

(https://en.wikipedia.org/wiki/Partial_trace)

Extreme example: Subsystem of the Bell state

The Bell state is a pure state, but the reduced density operator of the first qubit is a mixed state (the completely mixed state)

Stronger entanglement (with env.) → More error (on the system)

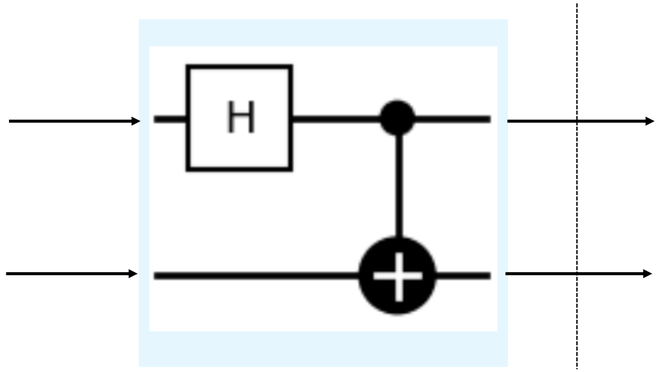
Bell state

$$|\varphi'_{se}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

System (Qubit) $|0\rangle$

$\otimes$

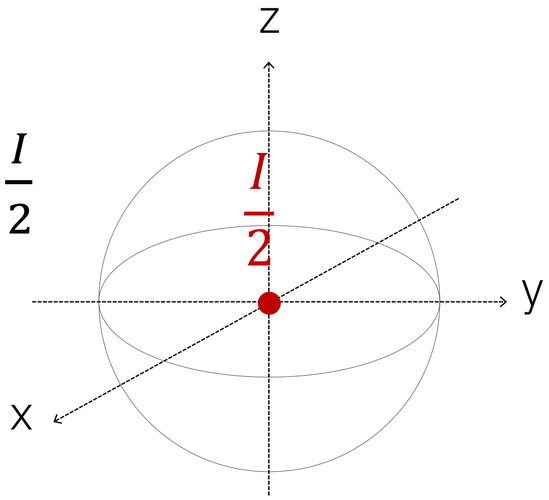
Environment (Qubit) $|0\rangle$



Quantum interaction as a whole

$$\rho'_s = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}$$

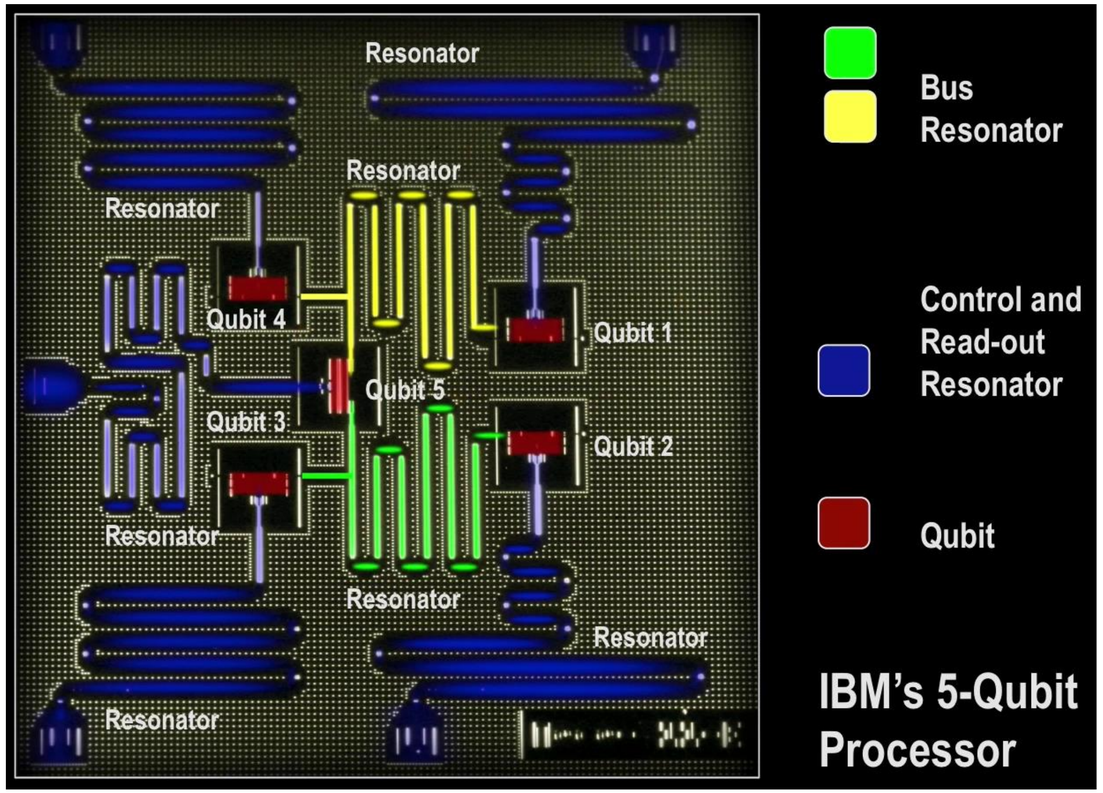
Completely mixed state



$$\rho'_s = \mathrm{tr}_e(|\varphi'_{se}\rangle\langle\varphi'_{se}|)$$

Partial trace over e (discard environment e, leave system s)

Measurement



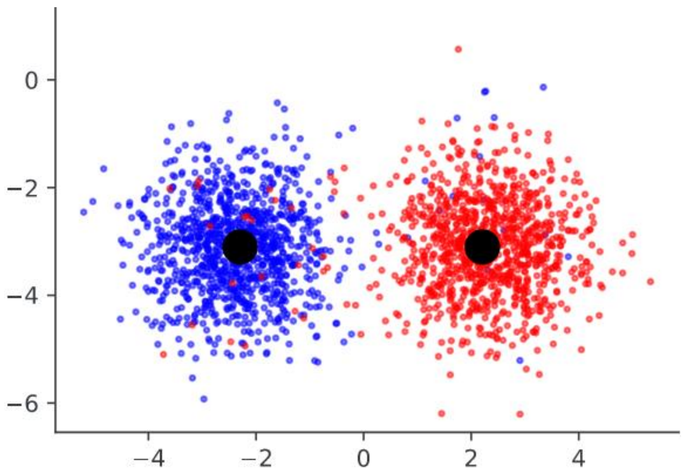
Measurement: a microwave readout pulse is sent into a readout resonator coupled to the qubit.

The response is a complex signal (with real and imaginary components) that can be captured as:

I: In-phase component (cosine part of the signal),
Q: Quadrature component (sine part of the signal)

Plot I + *i*Q

IQ Plane



Measurement or Readout error (SPA<u>M</u> error) (SPAM: State Preparation And **Measurement**)

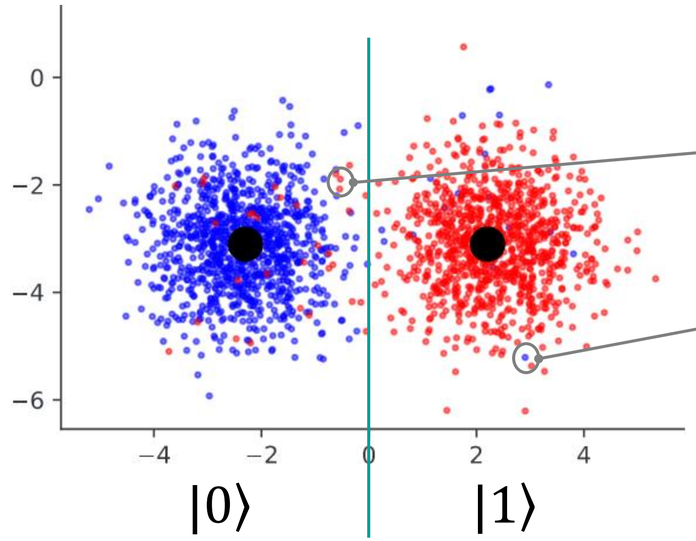
[Sources] • Failure to distinguish between the two states (during readout)

[Characters] • Classical errors (bit-flip errors)

[Measures] • (Better calibration), Error mitigation

Ex) IQ plot for qubit state readout

- Prepare $|0\rangle$
- Prepare $|1\rangle$



Two types of readout errors

$P(0|1)$: Prepare $|1\rangle$, Measure $|0\rangle$

$P(1|0)$: Prepare $|0\rangle$, Measure $|1\rangle$

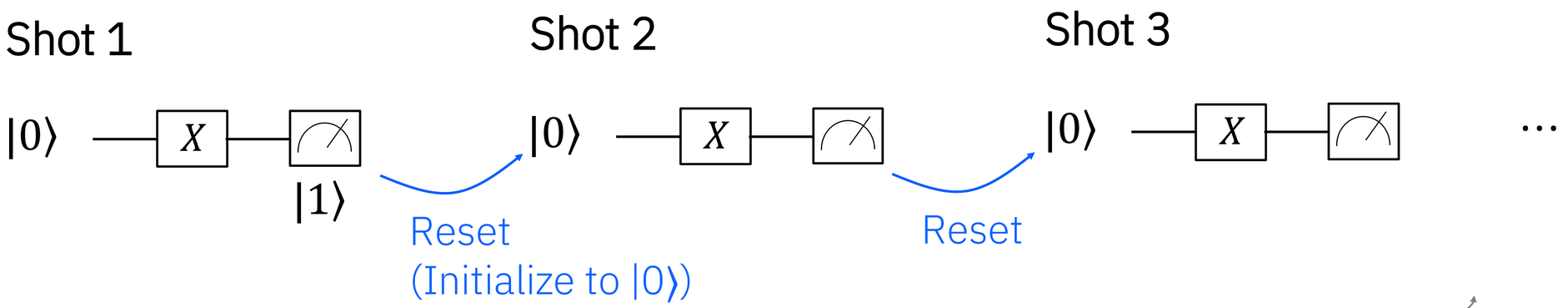
State Preparation Error (SPAM error)

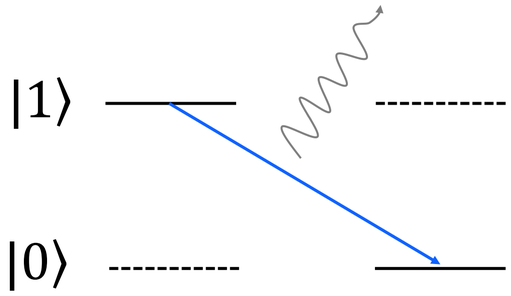
(SPAM: **State Preparation** And Measurement)

[Sources] • Imperfect reset (of a previously measured state)

[Measures] • Long shot intervals

Shots: Run a circuit multiple times to sample results (bits)





Sampling error (Shots error)

[Sources] • Core nature of quantum physics

[Measures] • Increase the number of shots

Measure a qubit → Observe a bit 0 or 1, following

Bernoulli distribution

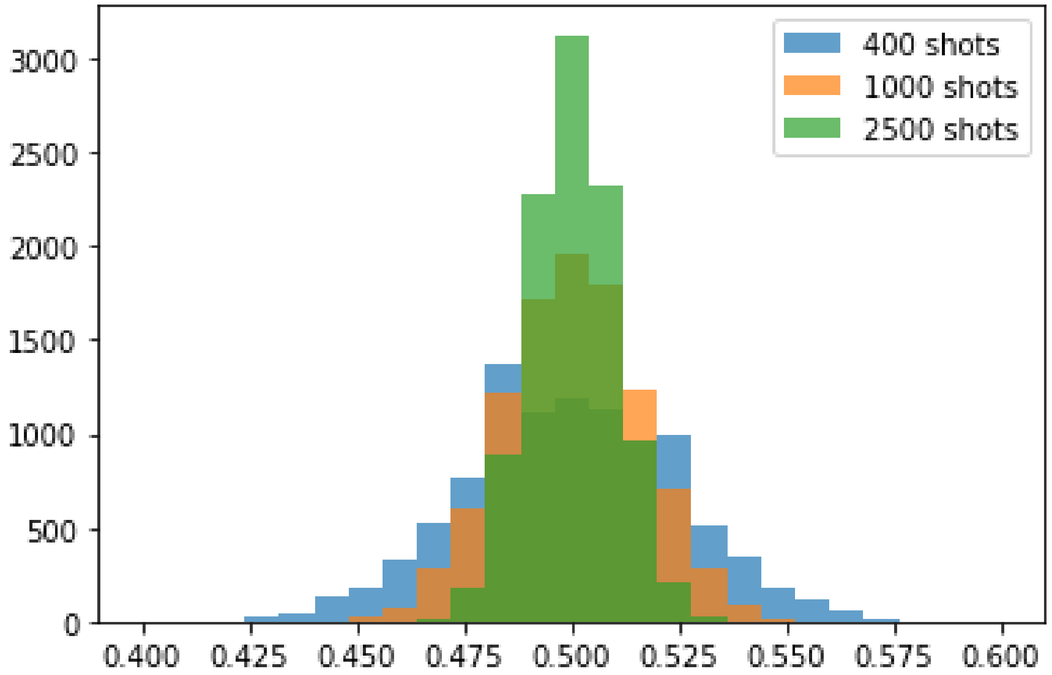
$$\begin{cases} 0 \text{ with probability } p \\ 1 \text{ with probability } 1-p \end{cases}$$

p depends on amplitude of $|0\rangle$ of the state

Measure multiple times to know p
→ Obtain sample mean of Bernoulli random variables $\hat{p}$

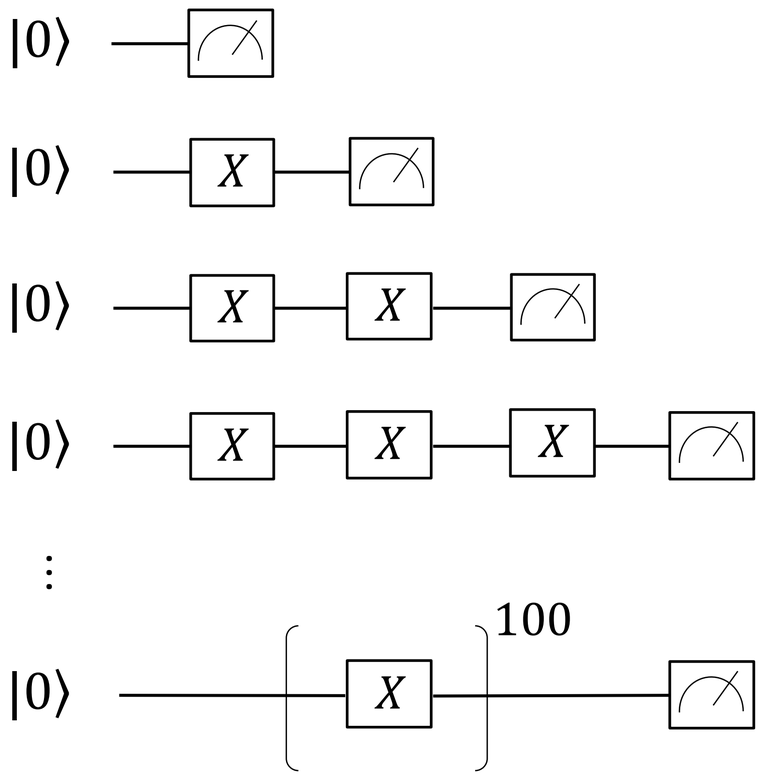
More shots → Less variance (more precise $\hat{p}$)

Distribution of mean of Bernoulli random variables (p=0.5)



Quiz: What errors look like

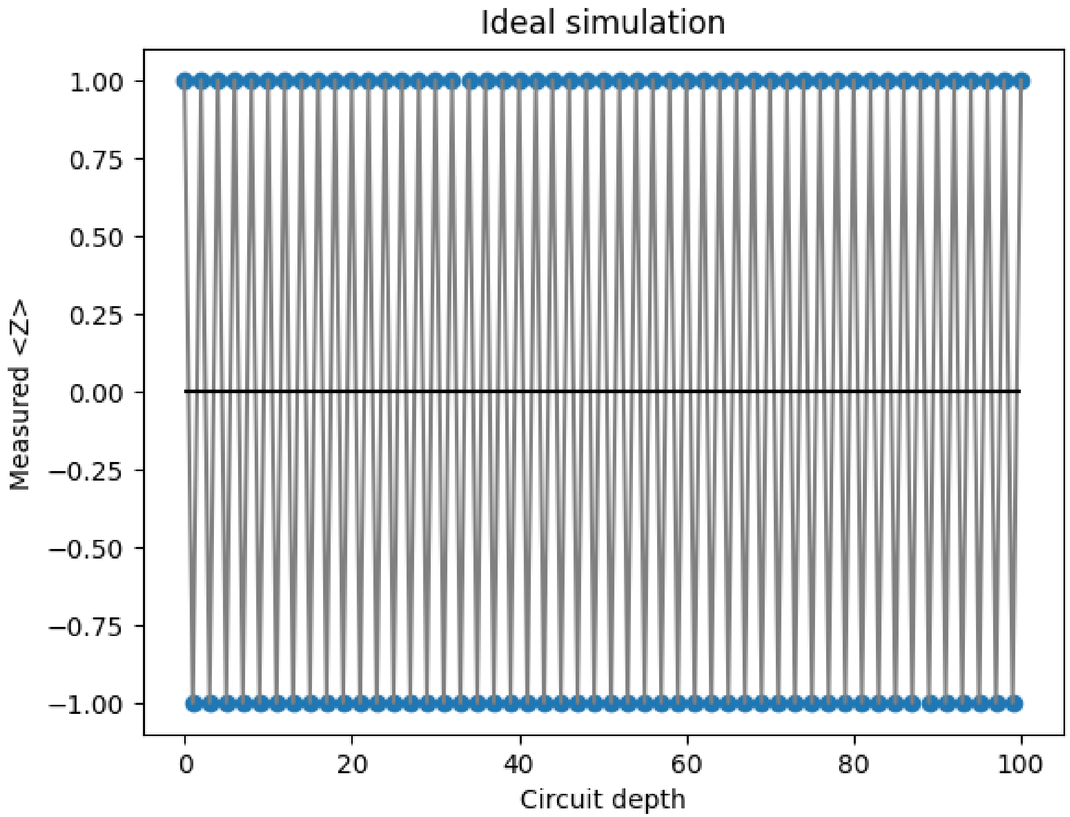
Run the following 101 circuits



400 shots for each circuit

Plot $\langle Z \rangle = \langle \varphi | Z | \varphi \rangle = P(0) - P(1)$

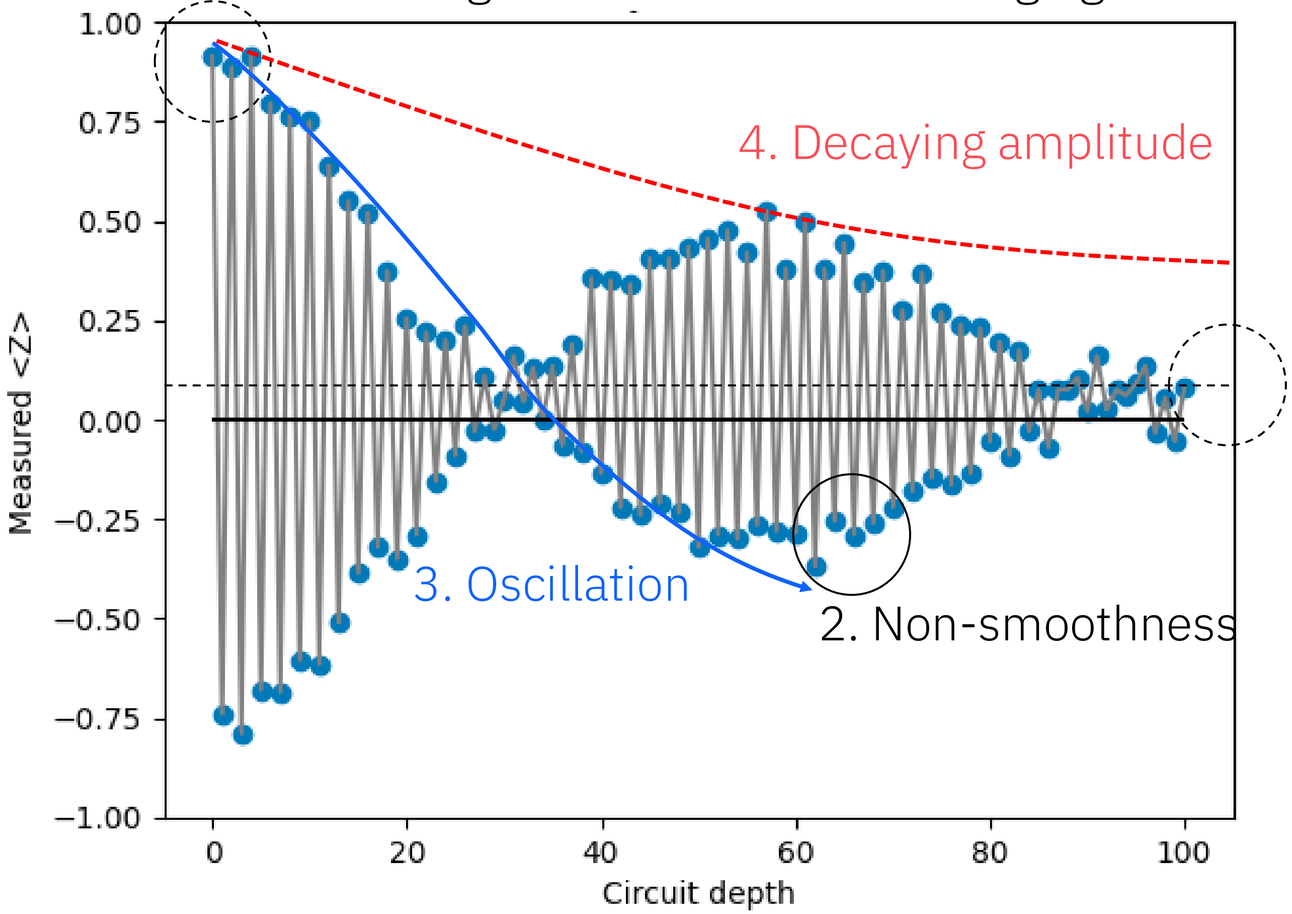
Ideally, observe 1 and -1 alternatively



Quiz: What errors look like

Running on noisy quantum computer, we observe

1. Not starting from 1.0 and not converging to 0.0



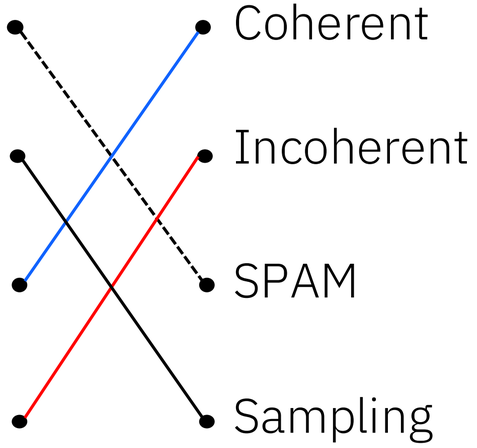
Why?

Connect an observation with the error causing it by a line

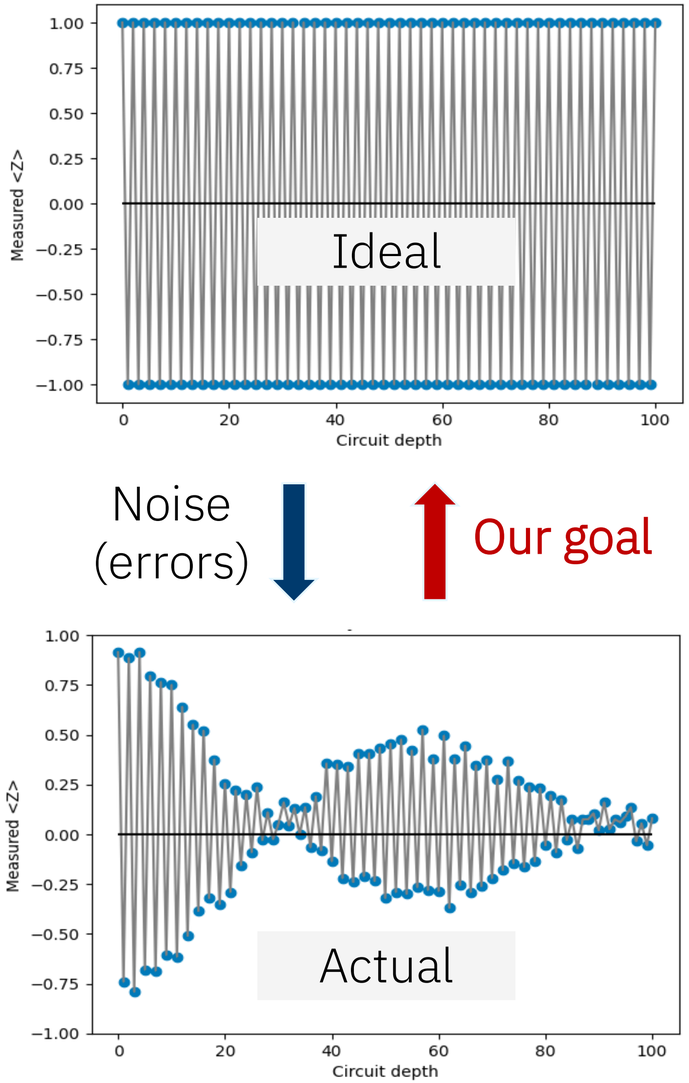
Observations:

1. Shrink/bias (d=0/∞)
2. Non-smoothness
3. Oscillation
4. Decay

Errors:



Fight gate/measurement errors



- Gate and measurement errors are dominant in today's superconducting-qubit computers

Classification of errors (CS view)	Affected operations:
• Coherent error	Gates
• Incoherent error	Gates
• SPAM error	Resets & measurements
• Sampling error	Measurements

(SPAM: State Preparation And Measurement)

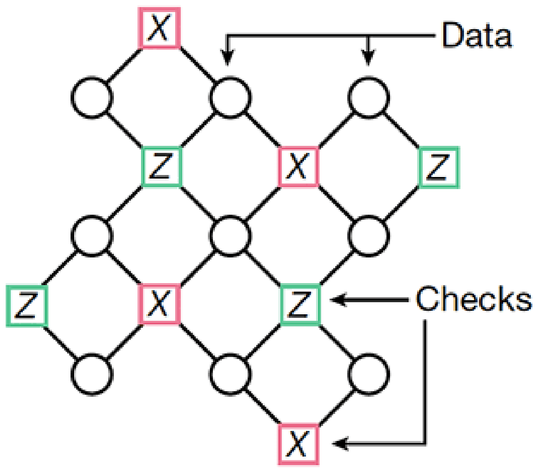
What you will learn today

- Talk (30min)
  - What is quantum noise/error
  - **Error suppression and mitigation techniques**
    - TREX (Twirled Readout Error eXtinction)
    - ZNE (Zero Noise Extrapolation)
    - PEA (Probabilistic Error Amplification)
- Break
- Hands-on (20 min)
- Theory (Hard – 30min)
  - Formalism of quantum errors
    - Standard error channels, e.g. Pauli error channel
    - Quantum channel
    - PTM (Pauli Transfer Matrix) representation

Error correction or error mitigation?

How to deal with errors due to noise?

Quantum error correction (QEC)

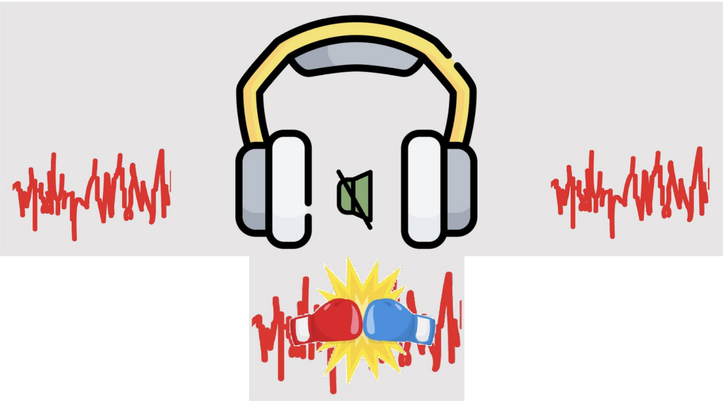


Source: Fig. 1 in [1]

Monitor
Error occurs
Error detected

Correct in quantum computation (in real time)

Quantum error mitigation (QEM)



Source: [2]

No monitor
Error occurs
Error undetected

Estimate corrected with classical computation (by post processing)

[1] Bravyi, S., Cross, A.W., Gambetta, J.M. *et al.* High-threshold and low-overhead fault-tolerant quantum memory. *Nature* **627**, 778–782 (2024).
[2] Minov Z., Probabilistic Error Cancellation with Sparse Pauli-Lindblad Models on Noisy Quantum Processors (https://www.youtube.com/watch?v=oPSBivh2rxQ)

Noise canceling headphones

Noise-Canceling Headphones

Goal: Remove background noise so you hear *just the music or voice*.

How it works:
Microphones pick up ambient
The headphones generate an **inverse sound wave** (same amplitude, opposite phase).
This wave **interferes destructively** with the noise → reducing or canceling it.

Result: cleaner signal for your ears.

Quantum Error Mitigation (QEM)

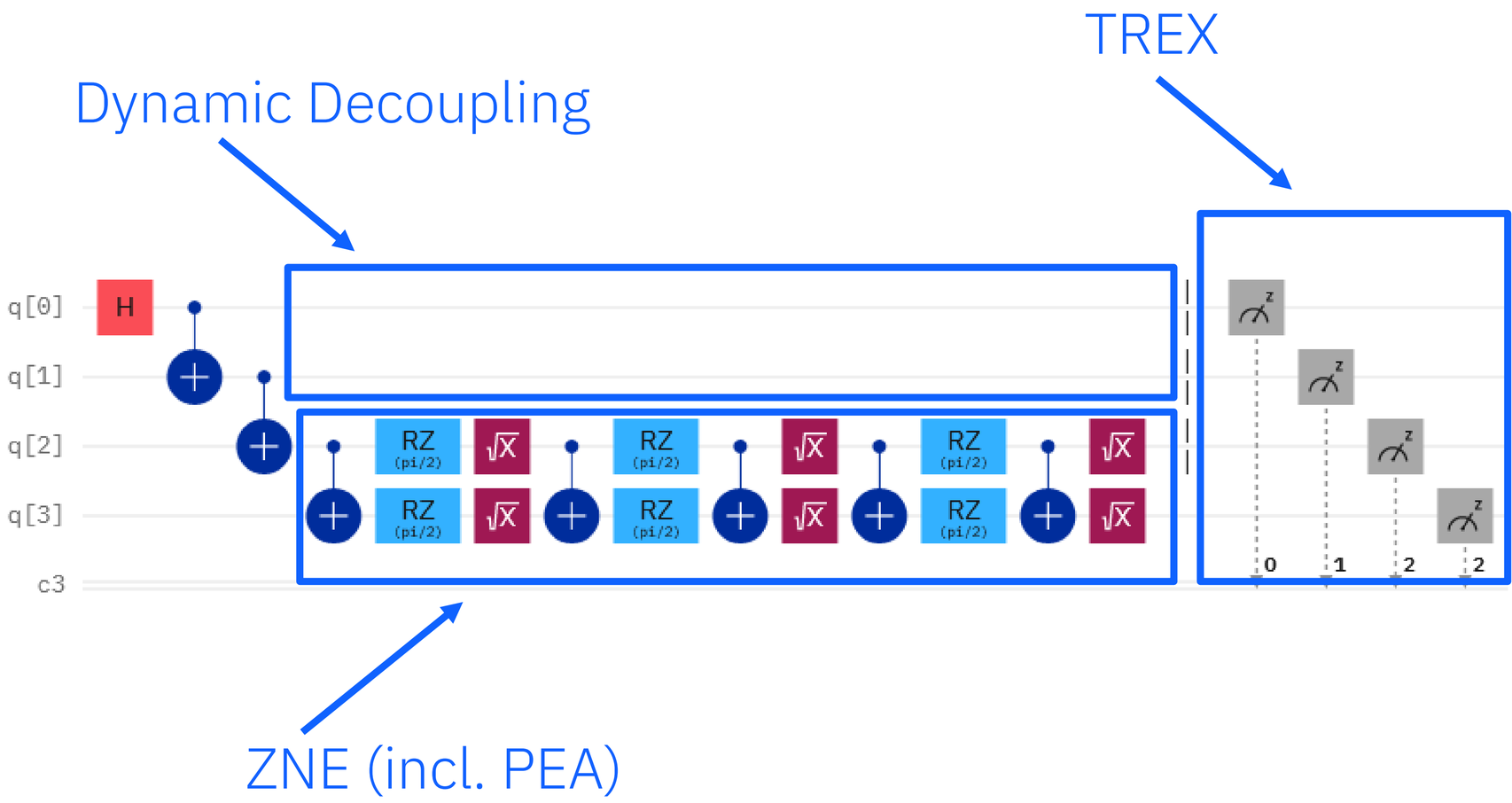
Goal: Reduce the effect of noise on quantum computations to get results that are **closer to the ideal, noiseless output**

How it works:
Measure noise (like measuring ambient noise).
Post-process the output data using classical algorithms to **undo the noise's effect**.

Result:

Error suppression and mitigation techniques

- Different types of errors need different suppression and mitigation techniques.
- Different types of techniques can be combined!



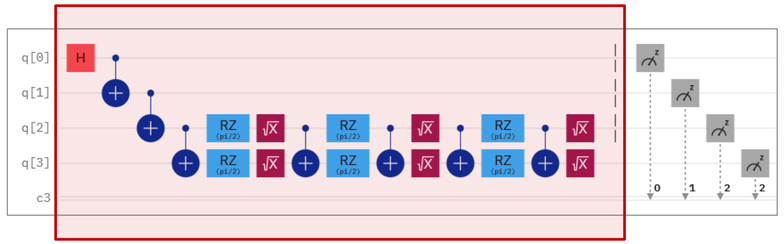
- Dynamic Decoupling
- TREX (Twirled Readout Error eXtinction)
- ZNE (Zero Noise Extrapolation)
- PEA (Probabilistic Error Amplification)

Error suppression and Error mitigation

Error suppression

Aim to reduce the error itself (in real time)

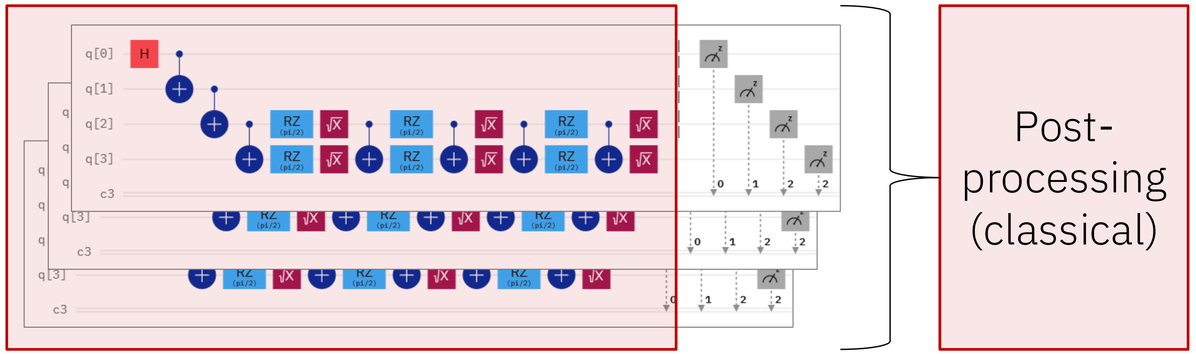
- Do something before measurement
- No change in the number of circuits
- Work even for a single shot

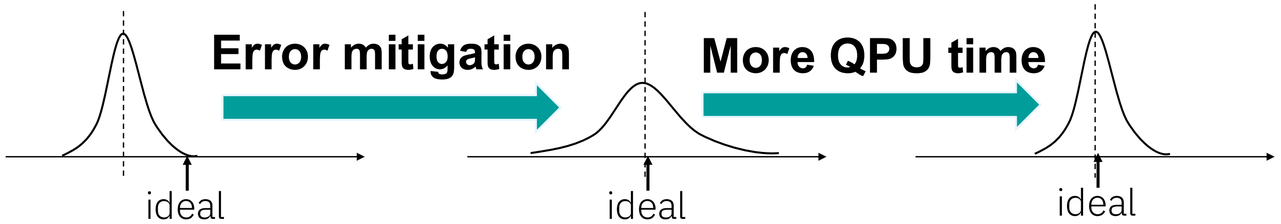


Error mitigation

Aim to recover the error-free result (with post-processing)

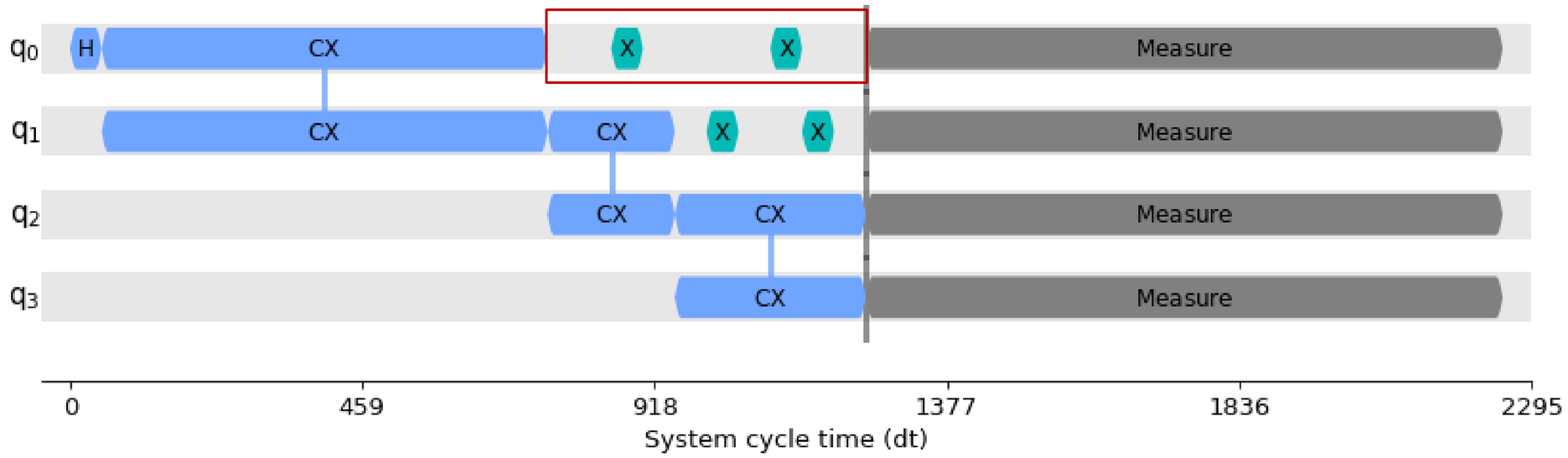
- Require classical post-processing
- Require more circuits to run
- Require multiple shots





Error suppression: Dynamical Decoupling (DD)

- **Suppress errors in qubit idling time** effectively
- Insert gates so it adds up to become an identity, e.g. X-X, X-Y-X-Y



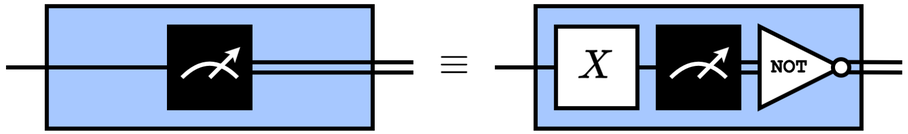
In this case, coherent Rz errors are cancelled out (assuming no errors on X gates):

$$R_z(\theta)\, X\, R_z(2\theta)\, X\, R_z(\theta) = R_z(\theta)\, R_z(-2\theta)\, R_z(\theta) = I$$

No DD → $R_z(4\theta)$ error

Twirled Readout Error eXtinction (TREX)

EV: Expectation Value

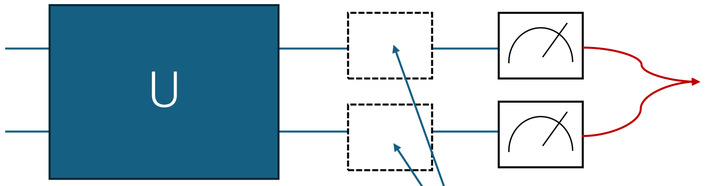
Idea: 

Focus on the computation of the EVs of Pauli observables composed only of Pauli I and Z for simplicity

Ex) EV of ZZ for $|\varphi\rangle = \mathrm{U}|00\rangle$, i.e. $\langle\varphi|ZZ|\varphi\rangle$

$$\langle ZZ\rangle = P(00) - P(01) - P(10) + P(11)$$

1. Original circuit (with random bit flipping)

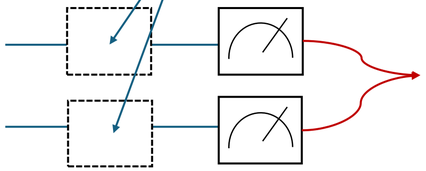


Flip measured bit if X gate is inserted → $\langle\tilde{\varphi}|ZZ|\tilde{\varphi}\rangle$

Randomly insert X gate

2. Calibration circuit

(Identity circuit with random bit flipping)



Flip measured bit if X gate is inserted → $\langle\tilde{0}|ZZ|\tilde{0}\rangle$

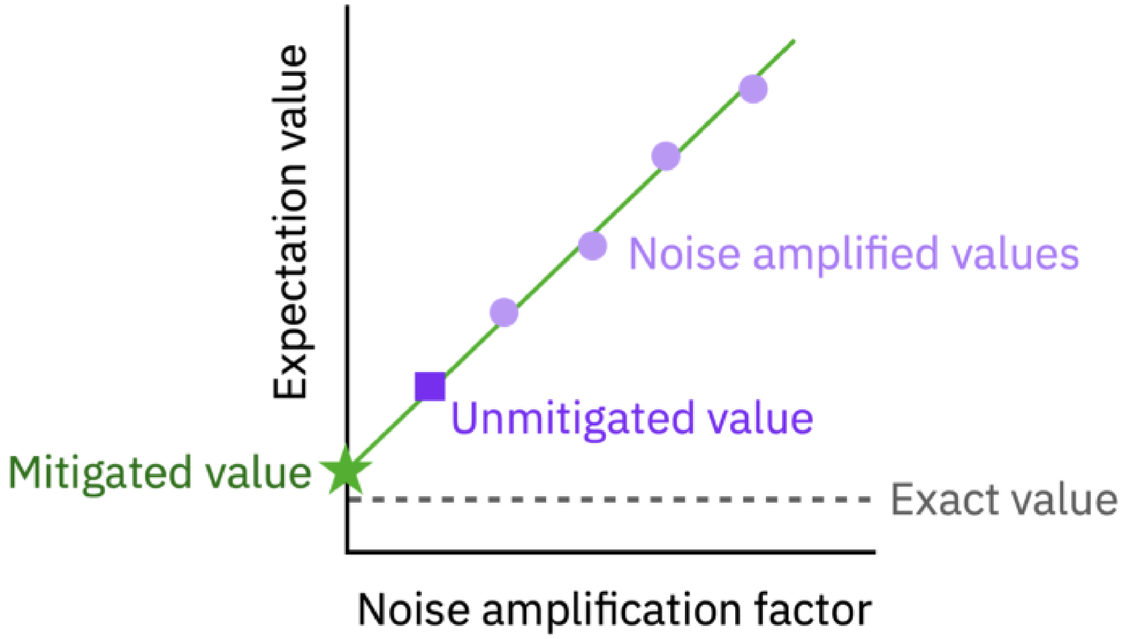
Mitigated EV = [EV from 1] / [EV from 2]

$\langle\tilde{\varphi}|ZZ|\tilde{\varphi}\rangle$ $\langle\tilde{0}|ZZ|\tilde{0}\rangle$

Van Den Berg, E., Minev, Z. K., & Temme, K. (2022). Model-free readout-error mitigation for quantum expectation values. *Physical Review A*, 105(3), 032620.

Zero Noise Extrapolation (ZNE)

Run multiple circuits with different gate error rates and extrapolate the expectation value at zero-noise point



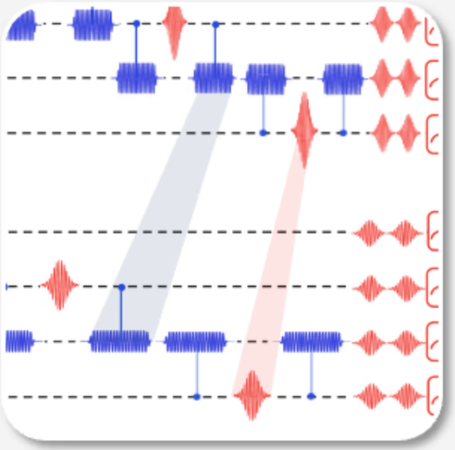
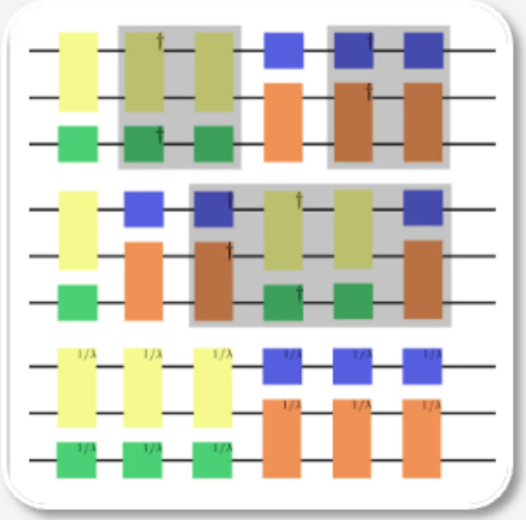
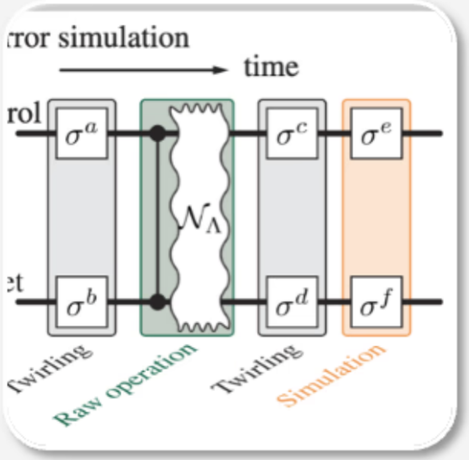
Options:

- Noise amplifier
- Noise factors e.g. [1.0, 1.2, 1.5], [1, 3, 5]...
- Extrapolator e.g. Linear, Quadratic, Exponential ...

https://github.com/Qiskit/qiskit-ibm-runtime/blob/stable/0.17/docs/tutorials/Error-Suppression-and-Error-Mitigation.ipynb

ZNE: Noise Amplification

- Pulse stretching assumes gate noise is proportional to duration, which is typically not true. Calibration is also costly.
- Gate folding requires large stretch factors that greatly limit the depth of circuits that can be run.
- PEA can be applied to any circuit that can be run with native noise factor (λ=1λ=1) but requires learning the noise model.
- You can write your own amplification!

Pulse Stretching	Gate Folding	Probabilistic Error Amplification
Scale pulse duration via calibration	Repeat gates in identity cycles $U \mapsto U(U^{-1}U)^{\lambda-1}/2$	Add noise via sampling Pauli channels
		
Kandala et al. Nature (2019)	Shultz et al. PRA (2022)	Li & Benjamin PRX (2017)

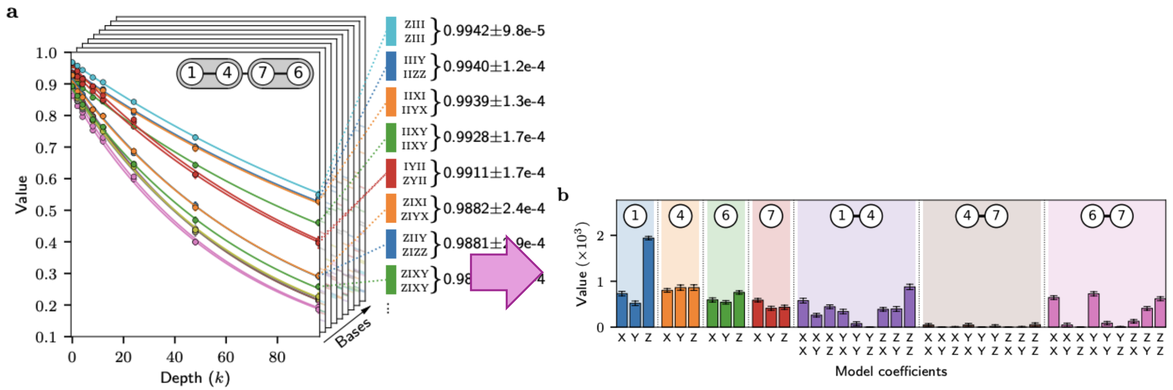
Probabilistic Error Amplification (PEA)

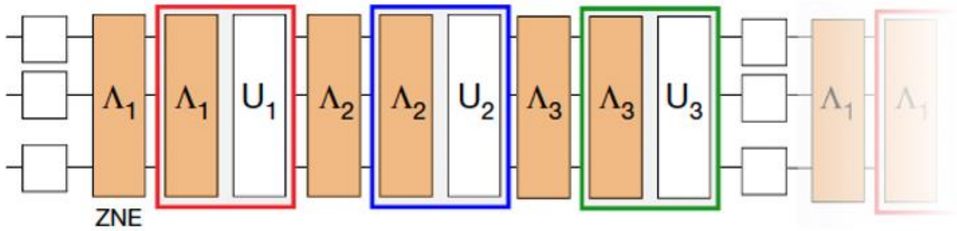
Pauli Twirling

1) Simplify noise: Gate noise → Pauli channel

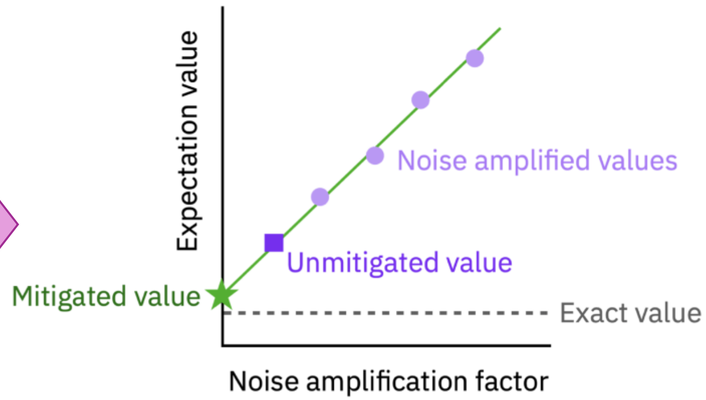
2) Learn noise (Estimate Pauli channel params)

3) Amplify noise + ZNE





Zero Noise Extrapolation (ZNE)



Kim, Y., Eddins, A., Anand, S., Wei, K. X., Van Den Berg, E., Rosenblatt, S., ... & Kandala, A. (2023). Evidence for the utility of quantum computing before fault tolerance. *Nature*, *618*(7965), 500-505.

Pauli Twirling

- Also called **randomized compiling**.
- Used to convert arbitrary noise channels into **Pauli channels**.
- Helps when dealing with coherent noise.
- Helps in the extrapolation stage of ZNE by making noise increase more or less monotonically.
- Often exclusively used on two qubit gates.

Clifford group

Pauli group

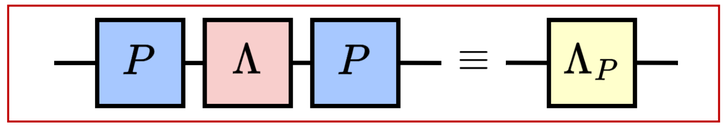
$$\mathbf{C}_n = \{V \in U_{2^n} \mid V\mathbf{P}_n V^\dagger = \mathbf{P}_n\}$$

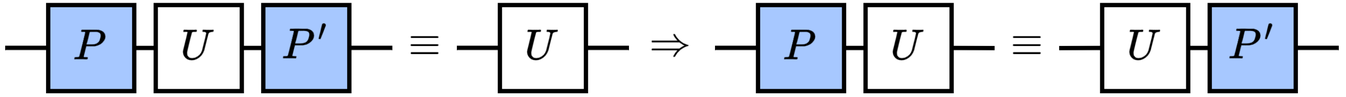
Clifford maps a Pauli to another Pauli by conjugation

$$U\,P\,U^\dagger = P'$$
$$\Leftrightarrow U\,P = P'U$$

Random Pauli P

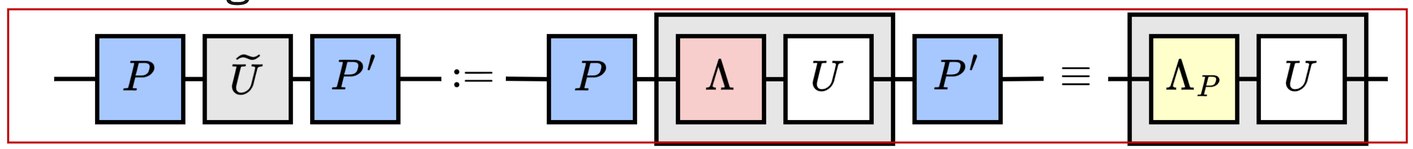
Goal:





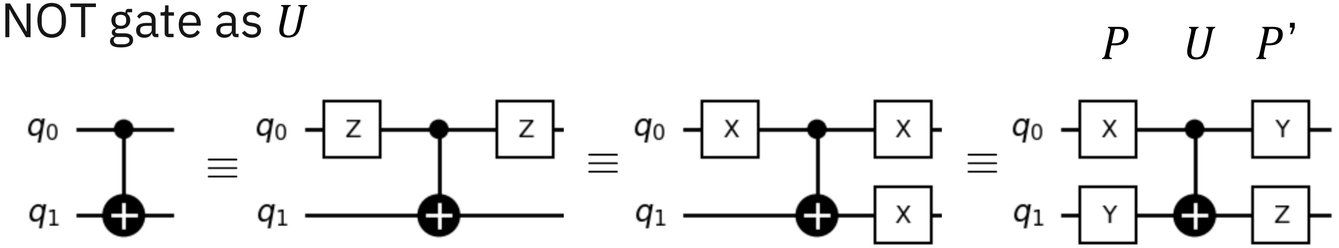
Clifford gate U

Implementation:



Partner Pauli P'

Ex) CNOT gate as U



What you will learn today

Part 1

- Noise characterization, error suppression and mitigation
  - What is quantum noise/error
  - Error suppression and mitigation techniques
    - TREX (Twirled Readout Error eXtinction)
    - ZNE (Zero Noise Extrapolation)
    - PEA (Probabilistic Error Amplification)

Part 2

- Hands-on coding session
  - Noisy simulation without error mitigation
  - More realistic noisy simulation
  - Real quantum computation with error mitigation

5 min break

- Please download the Jupyter notebook titled ‘**20260401_noise_characterization.ipynb**f’ from the github repo below.

https://github.com/qiskit-community/ntu-practitioner-26

References (Further reading)

- Introduction to Quantum Noise - Part 1 & 2 | Qiskit Global Summer School 2023
  Zlatko Minov
  - https://www.youtube.com/watch?v=3Ka11boCm1M
  - https://www.youtube.com/watch?v=gsKOx40gCUU
- Tensor networks and graphical calculus for open quantum systems
  Christopher J. Wood, Jacob D. Biamonte, David G. Cory
  - https://arxiv.org/abs/1111.6950
- A tutorial on tailoring quantum noise - Twirling 101
  Zlatko Minov
  - https://www.zlatko-minev.com/blog/twirling
- Exploring Quantum Channels | Understanding Quantum Information & Computation: Lesson 10
  John Watrous
  - https://www.youtube.com/watch?v=cMl-xIDSmXI

What you have learned today

- What is quantum noise/error

- Error mitigation techniques
  - TREX (Twirled Readout Error eXtinction)
  - ZNE (Zero Noise Extrapolation)
  - PEA (Probabilistic Error Amplification)

- Noise simulation with Qiskit Aer
- Applying error mitigation using Qiskit Estimator

Thank you

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